DRILL 25

ADVANCED FIRE SUPPRESSION

NFPA 1001 Level 2 (JPR 5.3.1) / NFPA 11

FOAM EDUCTORS & FIREFIGHTING FOAM APPLICATION

⏱ **Time:** 45–60 min | 🚒 **Apparatus:** Pumper / Engine | 🦺 **PPE:** Full Structural Turnout Gear & Gloves

👥 **Roles:** Drill Leader (Facilitator) | Safety Officer (Watcher) | Nozzle Team & Pump Operator

🎯 OBJECTIVE & RURAL FIREGROUND VALUE

Every firefighter must know how to properly rig an **In-Line Foam Eductor** to an attack line, set appropriate metering percentages for **Class A (wildland/structure)** and **Class B (flammable liquids)** foam concentrates, understand backpressure limitations, and apply foam using the three classic techniques: **Roll-On, Bank-Down**, and **Rain-Down**.

📦 TOOLS & APPARATUS SETUP

- Apparatus pumper with 1.75" (45mm) discharge.
- 95 GPM (or 125 GPM) in-line foam eductor with metering dial and pick-up tube.
- 150 to 200 feet of 1.75" hose line (maximum allowable length downstream of eductor).
- Matching 95 GPM (or 125 GPM) foam or combination fog nozzle.
- 5-gallon pail of training foam concentrate (or water with dish soap for zero-cost environmental drill).

⚠️ SAFETY WATCH-OUTS & STOP RULE

⚠️ **Foam Concentrate Exposure:** Foam concentrate is irritating to skin and eyes. Wear eye protection and structural gloves while handling concentrate pails and pickup tubes.

⚠️ **Environmental Runoff:** Ensure foam solutions do not drain directly into local salmon rivers, brooks, or storm sewers. Conduct drill over contained asphalt or gravel.

🛑 **THE STOP RULE: Shout "STOP!" immediately if the nozzle stream collapses or if foam concentrate spills uncontrollably.**

PHASE 1: THE PHYSICS OF THE IN-LINE EDUCTOR (15 MINUTES) — SLOW DRY RUN

1. **The Venturi Effect:**
   - Water passing through a restricted orifice creates a low-pressure vacuum that draws foam concentrate up the pick-up tube.
2. **The 4 Rules of Foam Eductor Success (Why Eductors Fail):**
   - **Rule 1: Inlet Pressure:** The pump operator must deliver **200 PSI** to the inlet of the eductor. Any less will not create sufficient vacuum.
   - **Rule 2: Matching Gallonage:** The nozzle GPM **MUST MATCH** the eductor GPM rating! (e.g., A 95 GPM eductor requires a 95 GPM nozzle setting. If the nozzle is set to 60 or 150 GPM, foam will not draft!).
   - **Rule 3: Maximum Hose Length:** Never exceed **200 feet (60m)** of 1.75" hose between the eductor and the nozzle. Excessive downstream friction causes backpressure that kills the Venturi vacuum.
   - **Rule 4: Elevation Limit:** Keep the nozzle within 10 to 15 feet of elevation relative to the eductor.

DRILL 25: PRACTICAL EVOLUTIONS & TAILGATE DEBRIEF

Advanced Fire Suppression • Practical Application

PHASE 2: METERING DIAL SETTINGS (10 MINUTES) — BUDDY COACHING

- **Class A Foam (Wood, Paper, Structural, Brush):**
- Set dial to **0.3% to 0.5%** for initial structural knockdown.
- Set dial to **0.1% to 0.2%** for deep penetration during overhaul.
- **Class B Hydrocarbons (Gasoline, Diesel, Heating Oil):**
- Set dial to **1% or 3%** (check manufacturer pail instructions).
- **Class B Polar Solvents (Ethanol, E85, Alcohol, Brake Fluid):**
- Requires AR-AFFF (Alcohol Resistant foam); set dial to **3% or 6%**.

PHASE 3: THE 3 APPLICATION TECHNIQUES (25 MINUTES) — TIMED / STRESS

1. **Technique 1: The Roll-On Method:**
   - Used on pool/spill fires on flat ground.
   - Direct the foam stream at the ground **just in front of the burning spill**.
   - The foam blanket builds up and rolls forward across the liquid surface, smothering vapors without agitating the fuel.
2. **Technique 2: The Bank-Down Method:**
   - Used when an obstacle or vertical wall is present (e.g., burning tanker trailer, wall, engine compartment).
   - Direct the stream against the vertical wall just above the liquid.
   - Foam bounces off the surface and runs gently down to cover the fuel pool.
3. **Technique 3: The Rain-Down Method:**
   - Used when roll-on or bank-down are obstructed.
   - Direct stream high into the air at a 60-degree angle over the fire.
   - Allow the foam to break apart and fall gently down like snowflakes over the spill surface (never plunge stream directly into fuel!).

⚠️ COMMON PITFALLS & FIREGROUND HAZARDS

- ❌ *Plunging the stream directly into the fuel:* Shooting a high-velocity stream into burning liquid splashes burning fuel across the ground and breaks up the foam blanket!
- ❌ *Partially closing the nozzle bail:* Gating or partially throttling the nozzle increases downstream backpressure, instantly halting foam induction at the eductor.
- ❌ *Kinked hose downstream:* Any kink creates backpressure that causes the eductor to stop picking up concentrate.

🗣️ 5-MINUTE TAILGATE DEBRIEF

1. ***"Why did the eductor stop pulling foam when the nozzle bail was half closed?", "Which application technique produces the gentlest foam blanket?"***

✅ APPARATUS RESET CHECKLIST

- ☐ Flush the eductor: Submerge pick-up tube in clean bucket of water and flow line for 60 seconds to wash corrosive foam out of the ball check valve.
- ☐ Hose drained, washed, and repacked.
- ☐ Concentrate pails sealed tightly and stowed upright in compartment.
- ☐ Attendance roster signed.

DRILL LEADER SIGNATURE	SAFETY OFFICER SIGNATURE	DATE COMPLETED	TRAINING HOURS